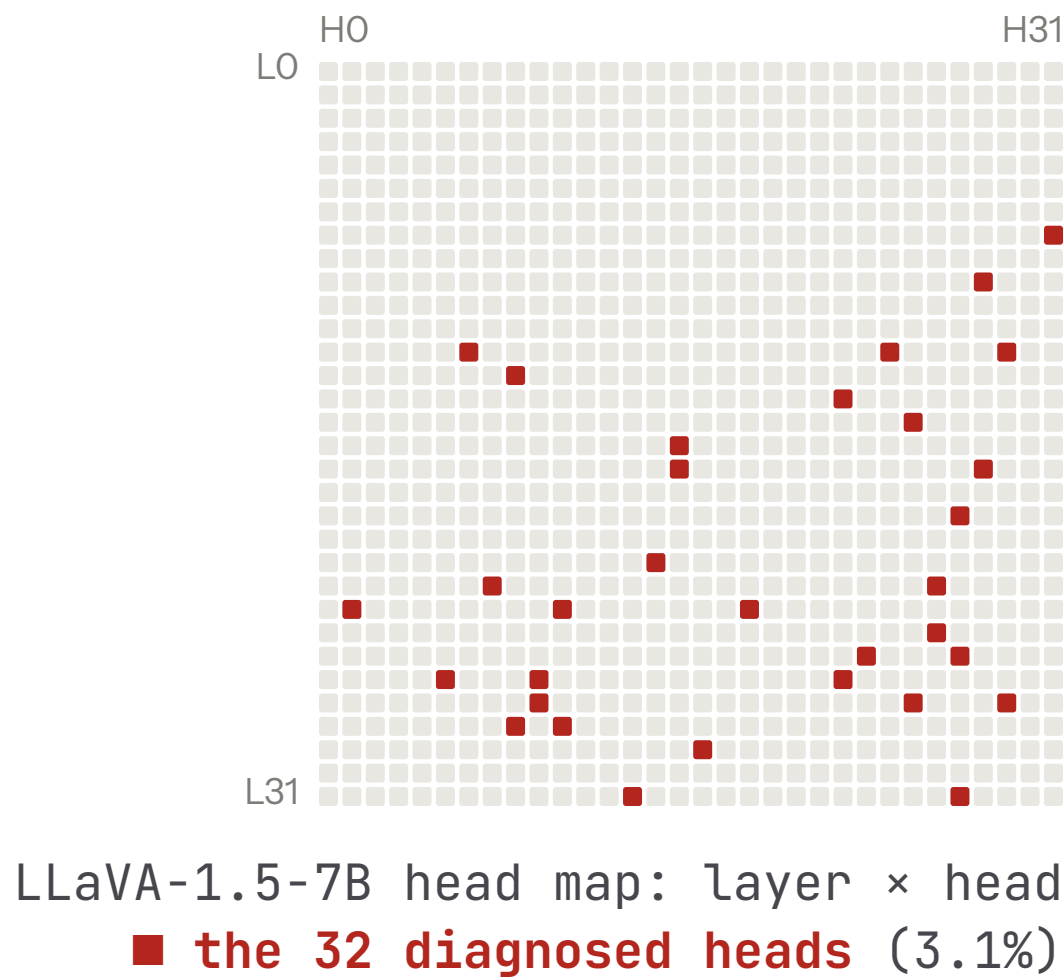


Targeting the Attention Heads Behind Object Hallucination in LLaVA

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Can an interpretability diagnosis tell us where to intervene, and what does acting on it actually change?

A diagnosis picks **32 of LLaVA's 1,024 heads**. Restricting a LoRA adapter and a decode-time guard to them cuts object hallucination **~38%** on held-out COCO; **layer-matched random heads do nothing**. The cost: shorter, more conservative captions.

0.370 → 0.230

CHAIRs, -38%
p < 0.001, 400 images

0.156 → 0.096

CHAIRi, -39%
p < 0.001, 400 images

0.400 vs 0.100

CHAIRs, random vs. diagnosed heads, same LoRA recipe

01 One diagnosis, two interventions

1 Diagnose · find the heads

Caption 200 COCO val2014 images; label each object mention **grounded** or **hallucinated**. Rank all 1,024 heads by how much their **image attention drops** at hallucinated vs. grounded object words.

Ablation screen: knock out candidates one at a time and keep the 32 with the largest effect on hallucination-token log-prob.

→ **32 heads (3.1%) across 19 of 32 layers** (head map, top right). Fixed once, then reused by both interventions.

2 Adapt · head-sliced LoRA

same 32 heads

LoRA on Q/K/V of the layers holding selected heads; **gradient masking** restricts learning to those heads' dimensions. DPO-style objective on contrastive caption pairs:

✓ **CHOSEN**
COCO ground-truth caption

✗ **REJECTED**
its own hallucinated caption

rank $r = 8$ · $\alpha = 16$ · dropout 0.05 · 3 epochs

3 Guard · decode-time controller

same 32 heads

Each step, the share of the heads' attention on image tokens:

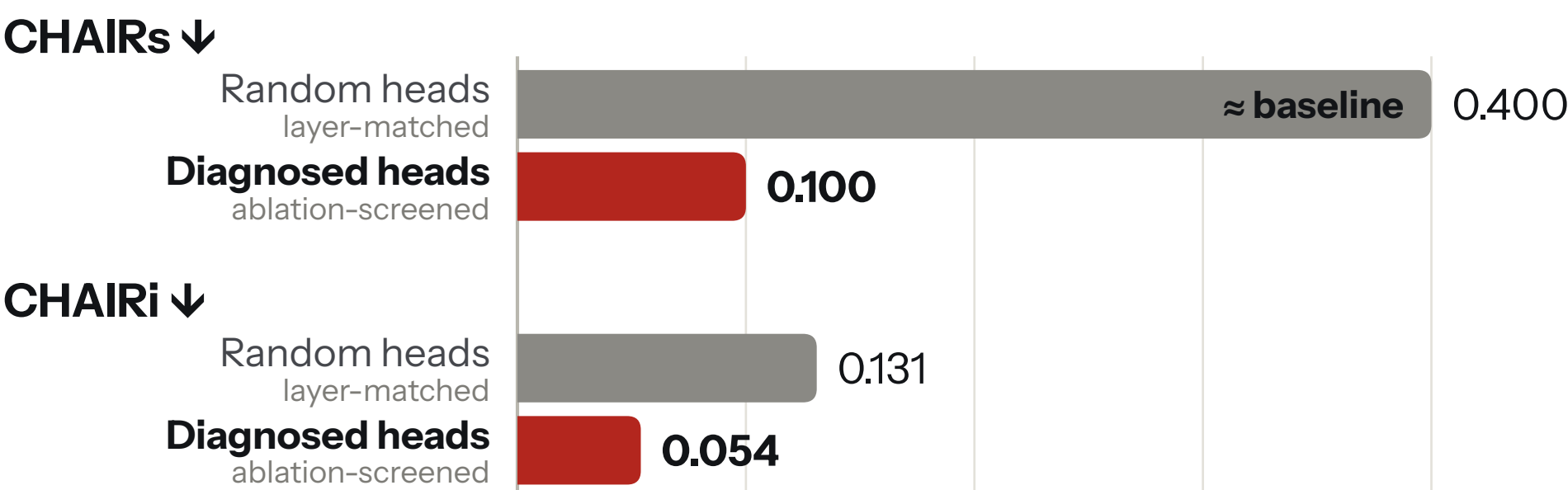
$$g_t = \text{mean}_{h \in H} \frac{\sum_{i \in \text{img}} \alpha_{h,i}}{\sum_j \alpha_{h,j}}$$

If the heads look away ($g_t < \theta$), penalize COCO object words v :

$$\ell_v \leftarrow \ell_v + \lambda \cdot \min(0, g_t - \theta)$$

$\theta = 0.08$ · $\lambda = 10$ · inactive while heads attend to the image

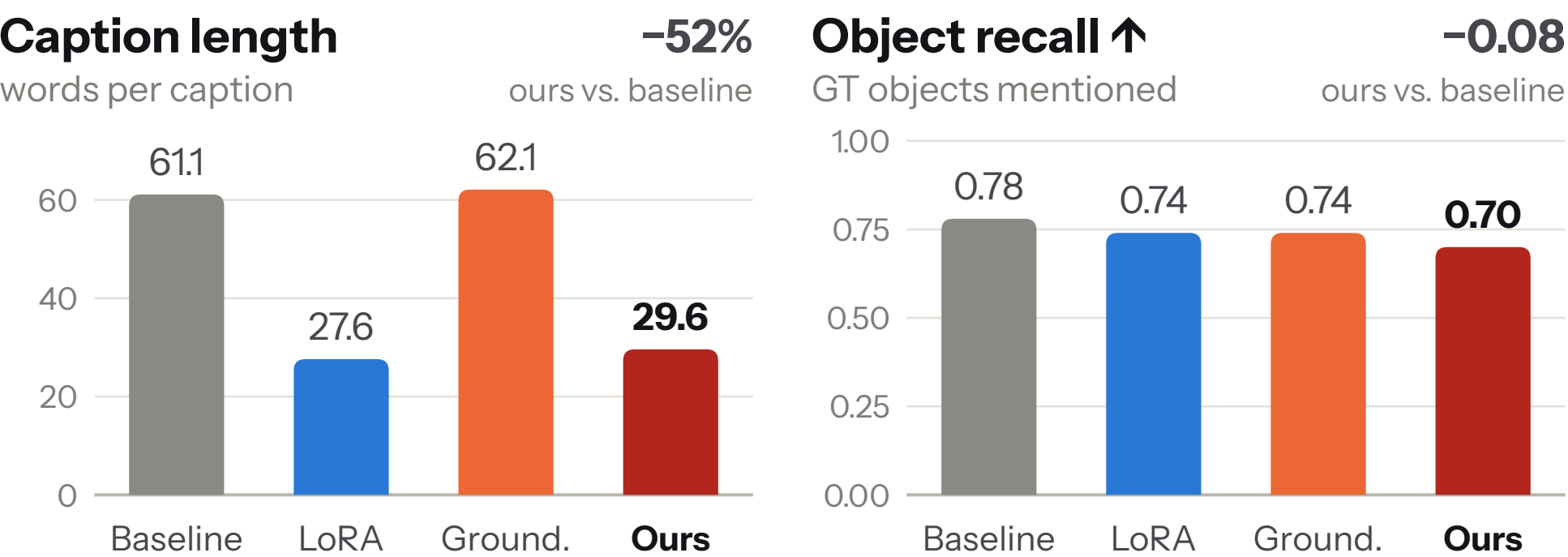
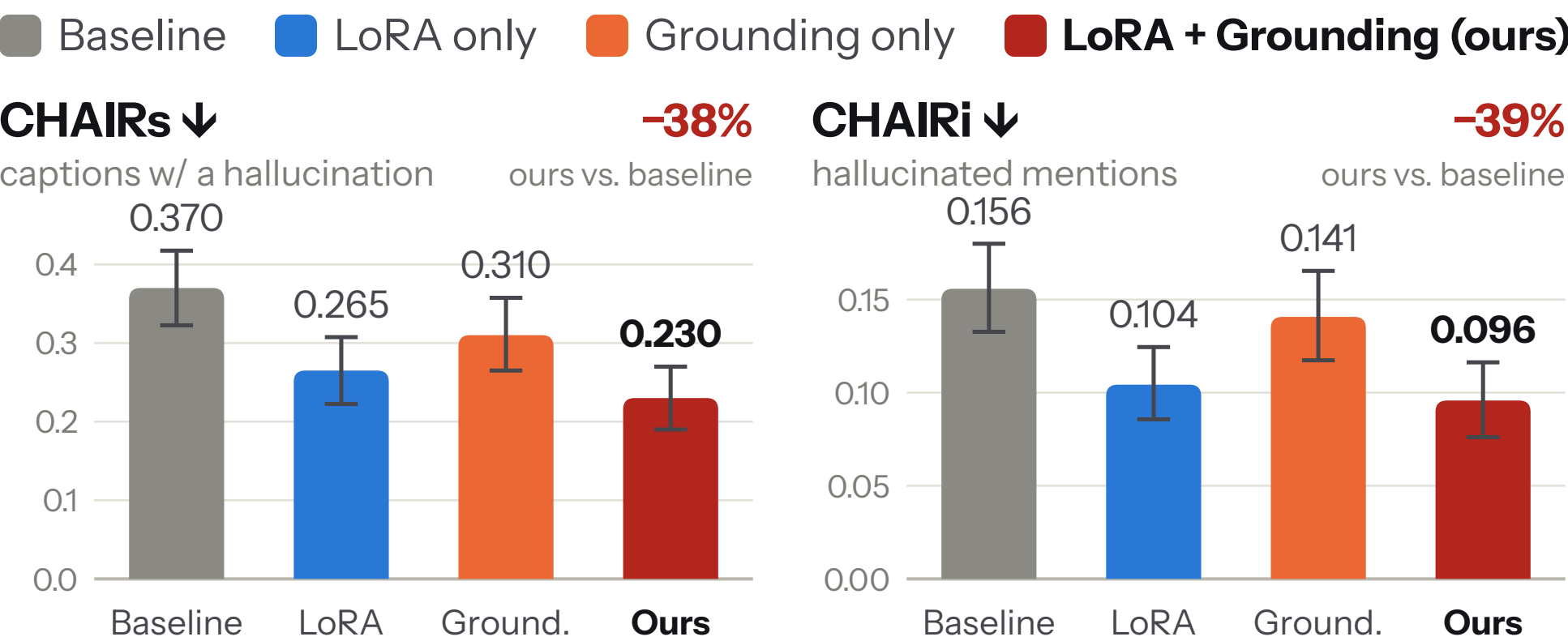
02 Selection matters, not capacity



Separate 200-image held-out split. Same DPO pairs, LoRA config and recipe; **only the head set changes**. Layer-matched random heads are indistinguishable from the baseline: the diagnosis finds sites with real leverage.

03 Fewer hallucinations, as a full profile

400 held-out COCO val2014 images, prompt “Describe this image in detail.”, greedy decoding. **CHAIRs**: share of captions with ≥ 1 hallucinated object. **CHAIRi**: share of object mentions that are hallucinated. Whiskers: bootstrap 95% CI.



Paired sign-flip tests vs. baseline: ours Δ CHAIRs **-0.140** [-0.190, -0.090], Δ CHAIRi **-0.060** [-0.084, -0.037], both p = 0.0001. Grounding alone does not significantly lower CHAIRi (p = 0.21): it is a modifier on LoRA, not a standalone fix.

Not a free lunch. Captions halve in length (61.1 → 29.6 words) and ground-truth object recall drops (0.78 → 0.70). CHAIR can fall just by saying less, so \$05 fixes the budget.

04 Example: held-out kitchen image



COCO objects: **refrigerator**, **sink**
green = grounded
red = hallucinated

Baseline
...A **refrigerator** is situated on the right side of the room, while an **oven** can be found near the center-left area.

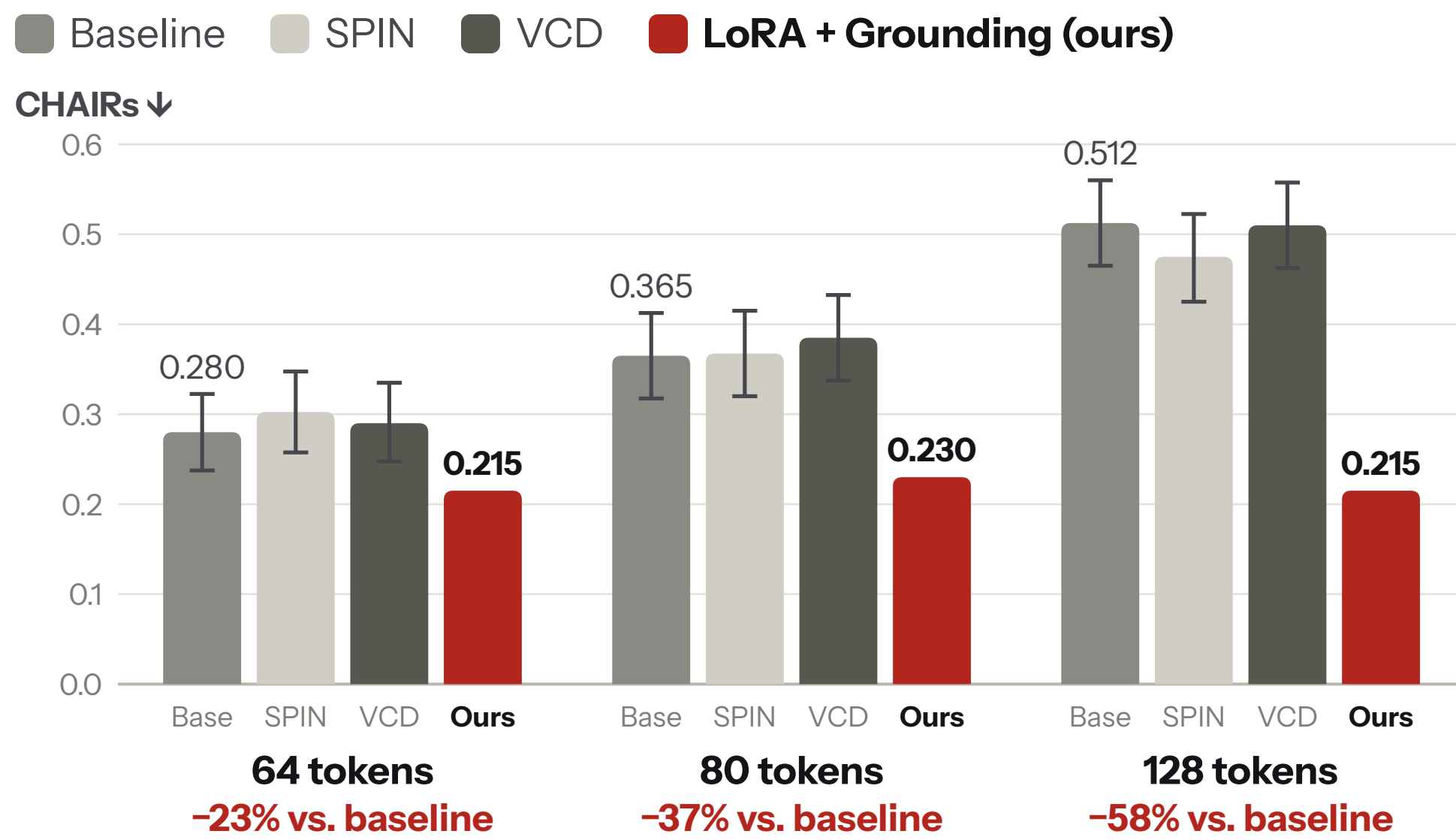
LoRA only
A kitchen with a **refrigerator** and **sink**. A brown counter top, cabinets and a window above the **sink**. A green **bottle** on the countertop.

Grounding only
...A **refrigerator** is situated on the right side of the room while an **oven** can be found in front of it. There are t...

LoRA + Grounding (ours)
A kitchen with a **refrigerator** and **sink**. A brown counter top in front of the **fridge**, cabinets above it, a window behind the stove topside down on its side.

Ours keeps the grounded objects and drops the oven, but is shorter and less fluent: the aggregate trade-off in miniature.

05 Not just shorter captions: fixed budgets

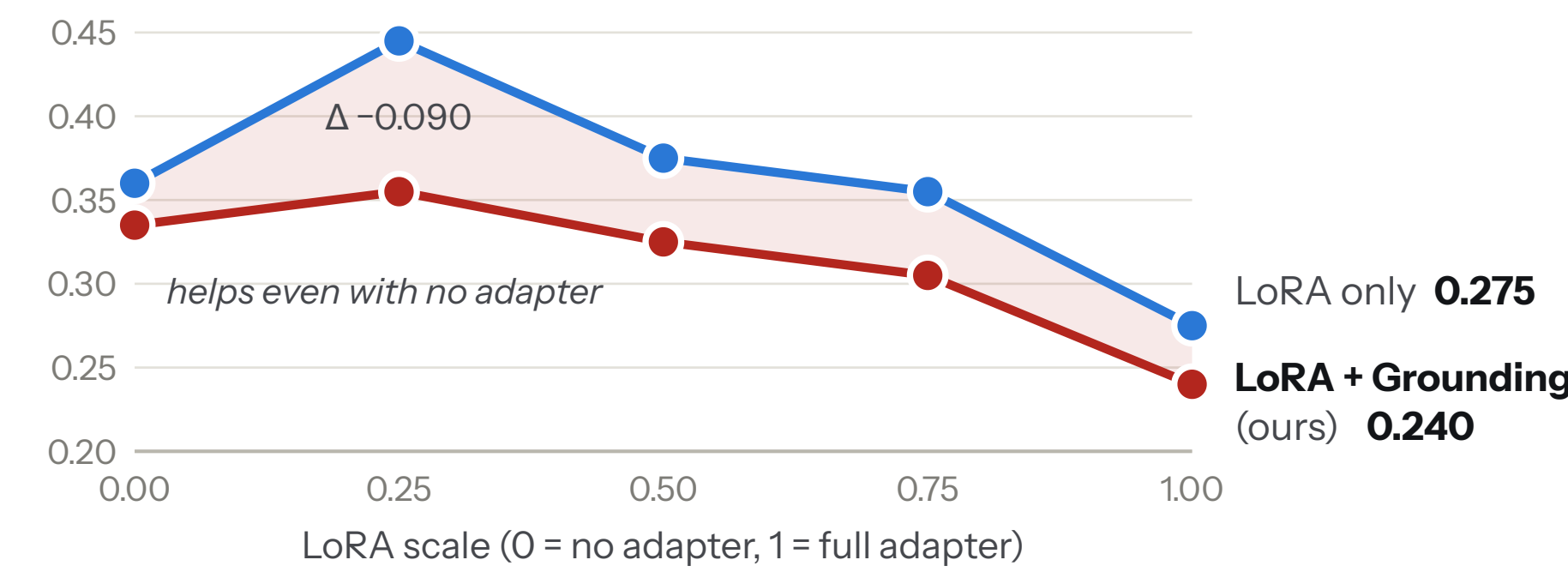


Same token budget for every method, same 400 images. Even at 64 tokens, where the baseline is truncated, ours cuts CHAIRs 23%; the gap grows to 58% at 128. SPIN and VCD do not consistently help here. **Caveat:** ours stays shorter within each budget (27.9 vs. 49.5 words at 64).

06 Sensitivity checks

Grounding helps at every LoRA strength

CHAIRs ↓ vs. adapter scale, 200 held-out images



0.24–0.26

CHAIRs for any threshold θ from 0.04 to 0.12

0.099 vs 0.098

CHAIRi, top-16 vs. top-32 heads (CHAIRs 0.24 both)

07 Takeaways

- 1 A **diagnostic signal** can choose **intervention sites** with real, non-random leverage.
- 2 One head set serves twice: **where to adapt** (LoRA) and **when to intervene** (decode-time guard).
- 3 Report a **behavioral profile**, not one score: hallucination, length, recall, budgets and baselines together.

Limits. One model, one prompt, COCO/CHAIR labels; attention is only a proxy for grounding. **Next:** length-matched comparison, all-head LoRA and plain-DPO baselines, human evaluation, more VLMs.